

**Learning Objective 3.9:** I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From weights to GUI settings.** A null-steering calculation for a 4-element subarray returns the complex weights

$$w_1 = 0.62 - j0.13, \quad w_2 = 0.88 + j0.05, \quad w_3 = 0.88 - j0.05, \quad w_4 = 0.62 + j0.13$$

The Phaser GUI takes element amplitude as a percentage of the largest weight and element phase as an offset in degrees.

- (a) Compute the four magnitudes and convert them to the percentages you would type into Element Gains.

gains =

- (b) Compute the four phase offsets in degrees.

phases =

- (c) A student enters the four gain percentages but forgets to enter the phase offsets. Describe the pattern they will measure and say whether a notch appears.

- (d) All four weights are multiplied by  $e^{j40^\circ}$  before being entered. How does the measured pattern change, and why?

2. **Reading a null-steered sweep.** A student sweeps the uniform 8-element array, freezes the trace, enters null-steering weights for a null at  $+22.5^\circ$ , and sweeps again. The two traces read as follows.

|                        | Uniform (frozen)   | Null steered |
|------------------------|--------------------|--------------|
| Peak level             | 0.0 dB (reference) | $-1.8$ dB    |
| Level at $+22.5^\circ$ | $-12.8$ dBc        | $-21.6$ dBc  |

- (a) How much additional rejection did the null steering buy at  $+22.5^\circ$ ?

$\Delta =$

- (b) Express the notch depth relative to the null-steered array's own peak, which is the number a jammer-to-signal calculation needs.

depth =

- (c) A second student reports a notch at  $-31$  dBc on the same bench with the same weights. The sweep's noise floor sits about 23 dB below the uniform peak. Is the report credible? Explain.

- (d) The notch appears centered at  $+19.7^\circ$  rather than  $+22.5^\circ$ . Give the most likely cause.

3. **Null depth under coarse quantization.** The lab is repeated on a board whose phase shifters have only  $B = 2$  bits.

(a) Find the phase LSB.

LSB =

(b) Use the rule of thumb that quantization sidelobes sit near  $-6B$  dB to estimate the deepest notch this board can hold.

depth  $\approx$

(c) Round the four phases from Question 1(b) to the nearest multiple of the  $90^\circ$  LSB. What does the resulting weight vector do?

phases =

(d) Explain why holding a deep null is more demanding of phase resolution than holding a beam peak.

4. **Two-channel MVDR by hand.** The PHASER's two digital channels are steered so that a target off boresight presents the real steering vector  $s = [1, -1]^T$ , while an interferer on boresight presents  $u = [1, 1]^T$  with power  $P = 9$  relative to a per-channel noise power of 1. The covariance seen by the beamformer is  $R = \sigma^2 I + P u u^T$ .

(a) Build  $R$ .

$$R = \boxed{\phantom{000000}}$$

(b) Compute the MVDR weights  $w = R^{-1} s / (s^T R^{-1} s)$ .

$$w = \boxed{\phantom{000000}}$$

(c) Evaluate the beamformer's response to the target,  $w^T s$ , and to the interferer,  $w^T u$ .

$$w^T s, w^T u = \boxed{\phantom{000000}}$$

(d) Which lab procedure did MVDR just reproduce, and what would  $w$  become if the interferer were switched off?

5. **Choosing an approach.** Each situation below uses the same 8-element PHASER.

(a) A fixed ground emitter sits at a surveyed bearing of  $+35^\circ$  from the array face and does not move. The array must keep its beam on a target at  $-10^\circ$ . Which approach would you use, and why?

(b) An airborne jammer of unknown and changing bearing appears during a mission. Which approach would you use, and why?

(c) A requirement calls for a 33 dB notch on a board whose coarse 3-bit phase shifters hold a null only to about 21 dB (Lesson 27). Taking roughly 6 dB of improvement per added bit, how many phase bits are needed?

$B =$

(d) Two jammers now illuminate the array from different bearings. State what happens when MVDR is asked to handle both on this hardware.

**Documentation:** \_\_\_\_\_